Dr.Varodom Toochinda
Dept. of Mechanical Engineering
Kasetsart University

Clustering Analysis

- 01211373 Machine Learning and Programming for Industry

Clustering Analysis

- Unsupervised learning
- Commonly-used algorithms
  - K-means (the only method covered in this lecture)
  - GMM (Gaussian Mixture Model)
  - EM (Expectation-Maximization)

K-means clustering algorithm

The k-means clustering algorithm is as follows:

1. Initialize **cluster centroids** $\mu_1, \mu_2, \ldots, \mu_k \in \mathbb{R}^d$ randomly.

2. Repeat until convergence: {

For every i set

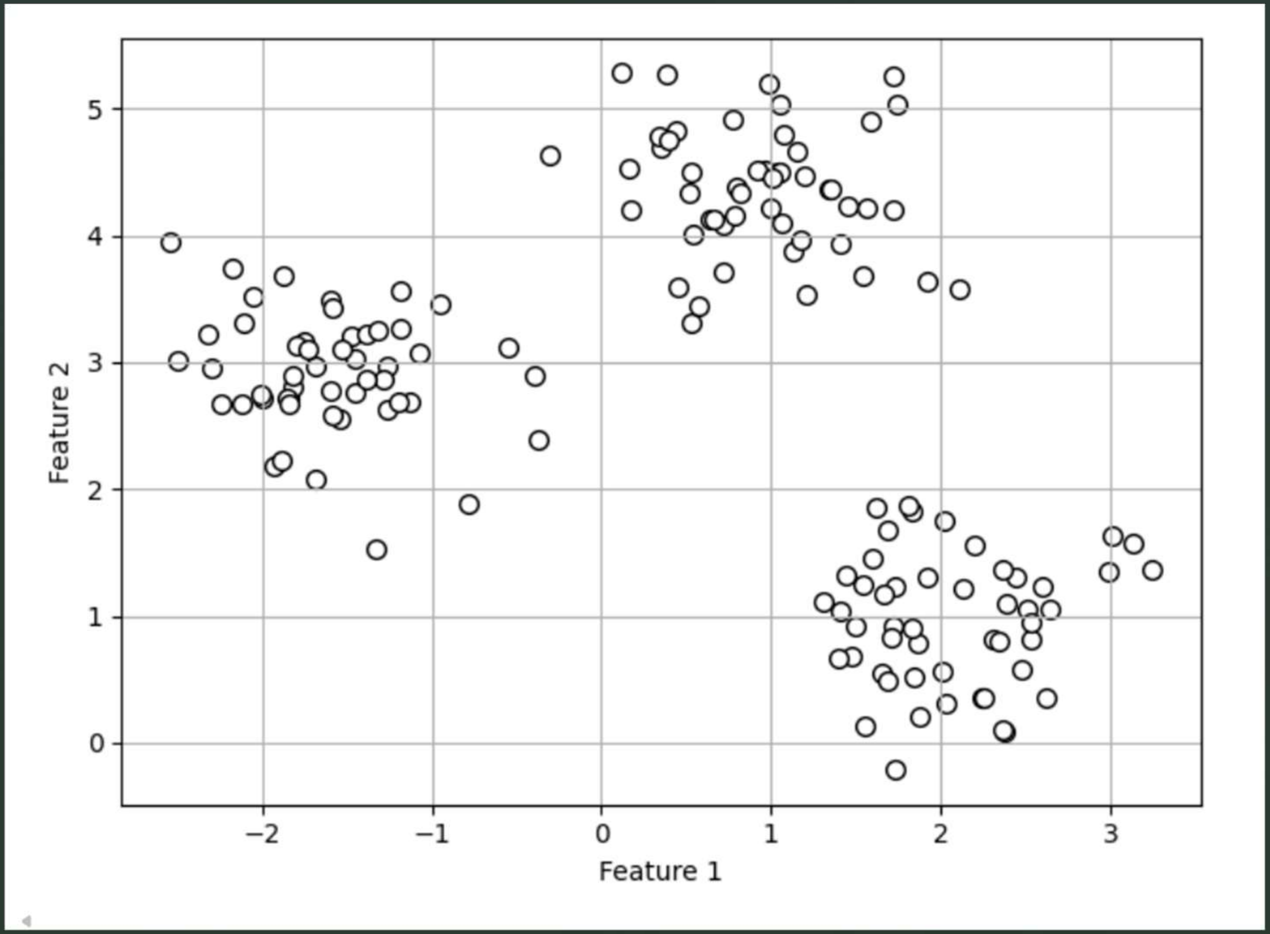
$$c^{(i)} := arg\ \underset{j}{min}\ \|x^{(i)} - \mu_j\|^2.$$

For each j set

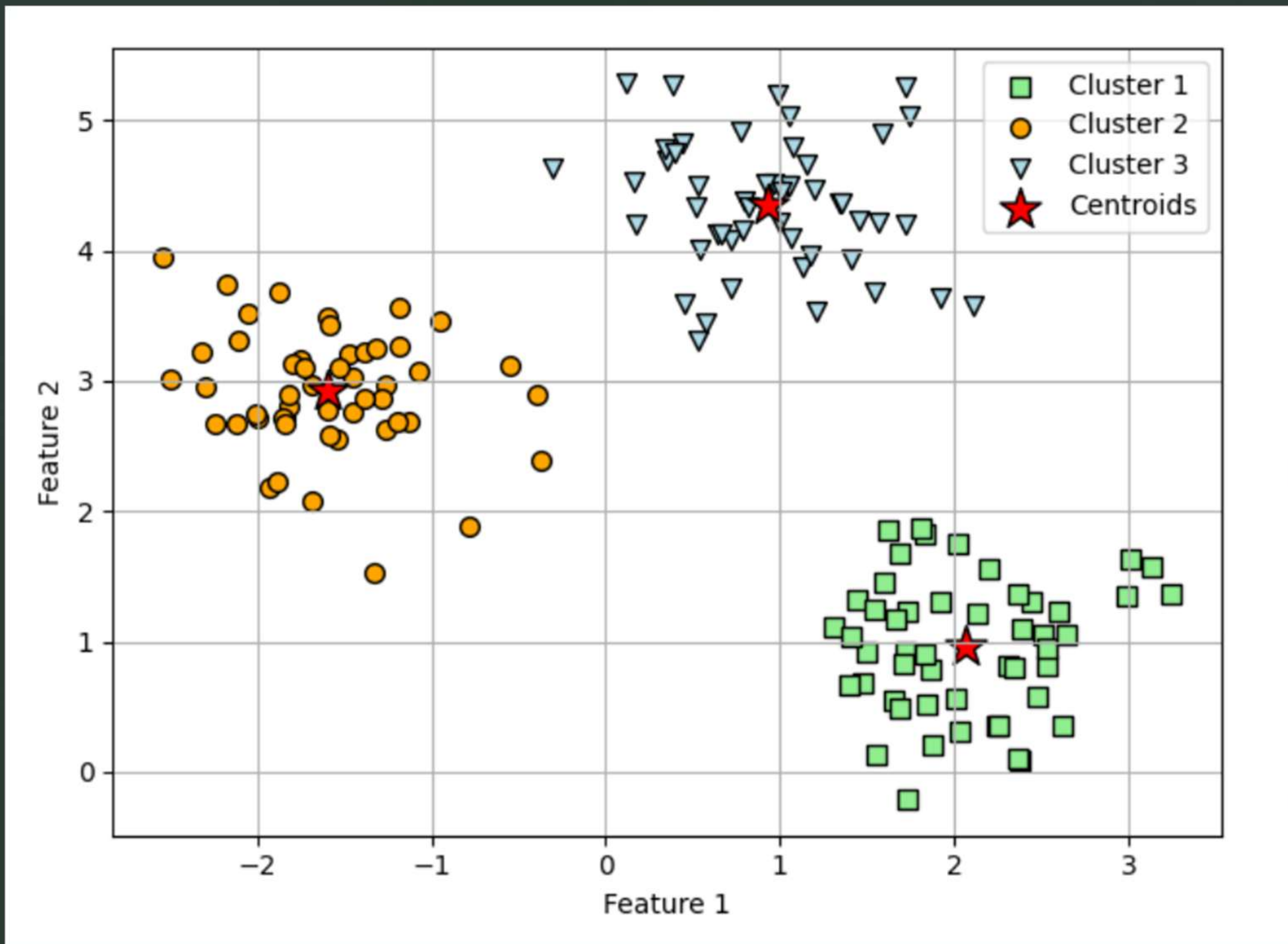
$$\mu_j := \frac{\sum_{i=1}^{n} 1\{c^{(i)} = j\} x^{(i)}}{\sum_{i=1}^{n} 1\{c^{(i)} = j\}}$$

}

Example : (k=3)

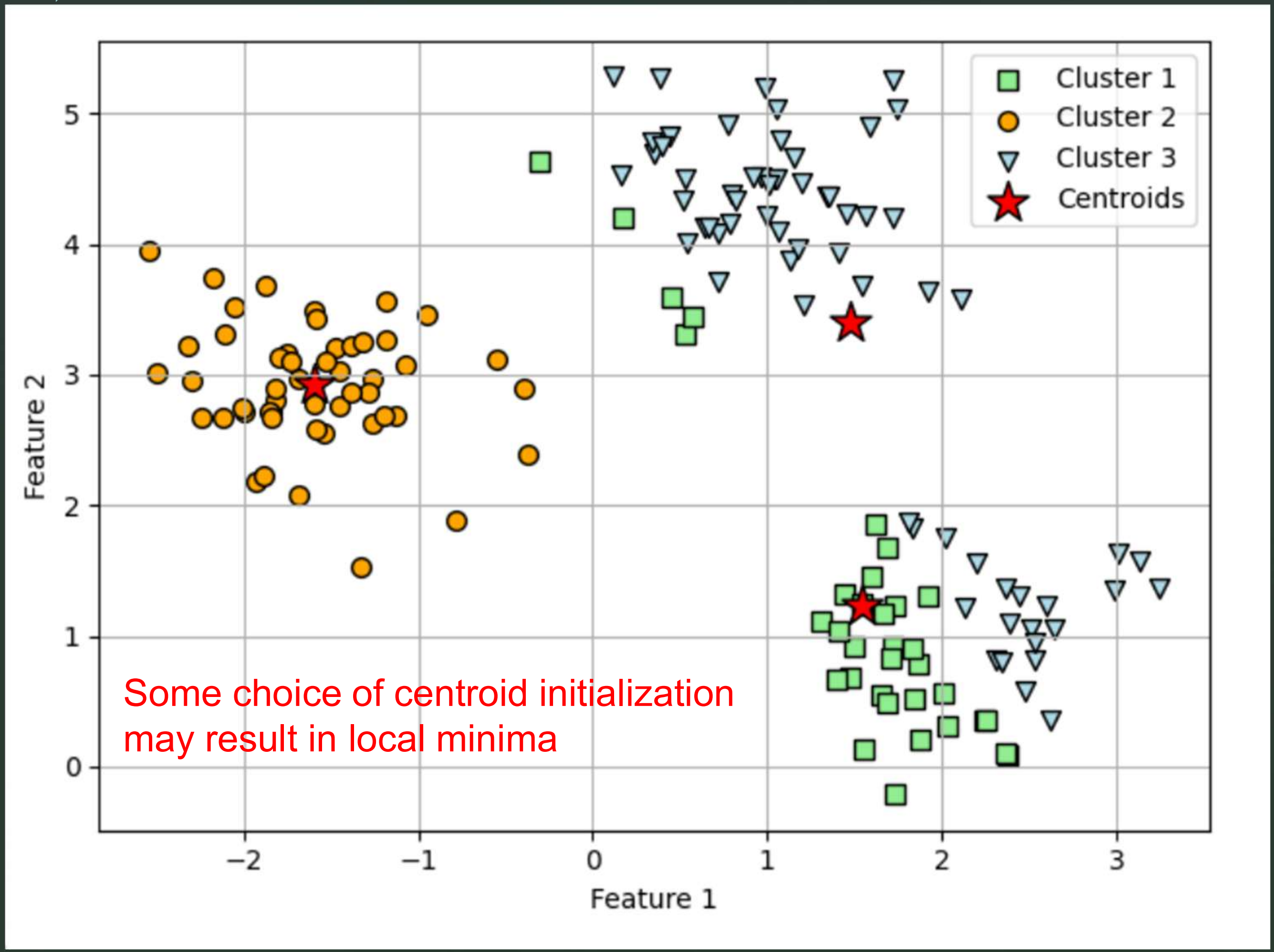


Example : (k=3)



Cluster 1
Cluster 2
Cluster 3
Centroids
Feature 1
Feature 2
−2
−1
0
1
2
3
4
5

Example : (k=3)


Cluster 1
Cluster 2
Cluster 3
Centroids
Feature 1
Feature 2
Some choice of centroid initialization
may result in local minima

K-means ++ algorithm (scikit-learn)

- Find some proper way to initialize the centroids by trying to spread them further apart
- Some choices may still result in local minima
- One can try a few choices of initialization and select the one that gives best result

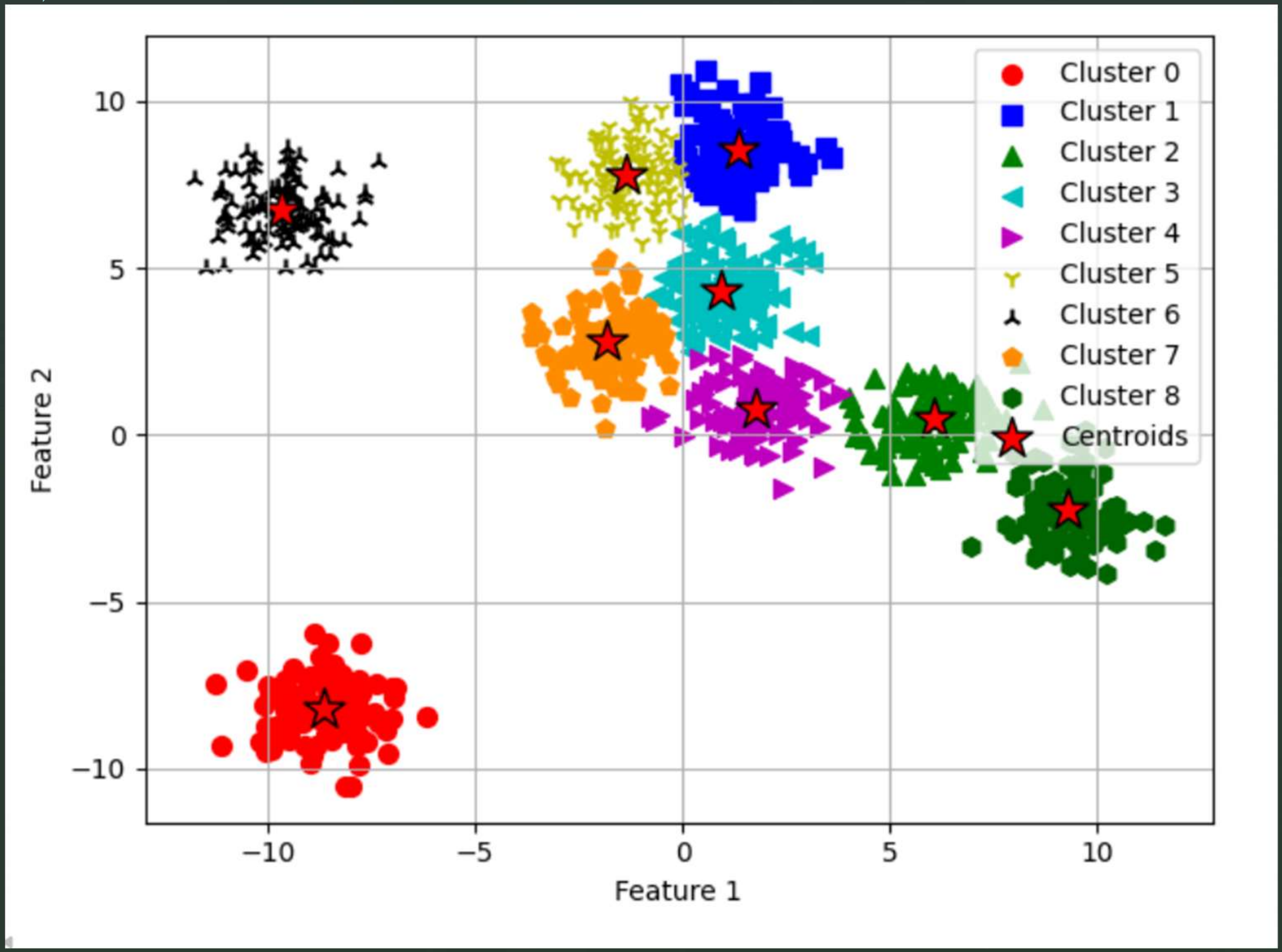
K-means++ algorithm

The initialization in k-means++ can be summarized as follows:

1. Initialize an empty set, $\mathbf{M}$, to store the k centroids being selected.
2. Randomly choose the first centroid, $\boldsymbol{\mu}^{(j)}$, from the input examples and assign it to $\mathbf{M}$.
3. For each example, $\boldsymbol{x}^{(i)}$, that is not in $\mathbf{M}$, find the minimum squared distance, $d(\boldsymbol{x}^{(i)}, \mathbf{M})^2$, to any of the centroids in $\mathbf{M}$.
4. To randomly select the next centroid, $\boldsymbol{\mu}^{(p)}$, use a weighted probability distribution equal to $\frac{d(\boldsymbol{\mu}^{(p)},\mathbf{M})^2}{\sum_i d(\boldsymbol{x}^{(i)},\mathbf{M})^2}$. For instance, we collect all points in an array and choose a weighted random sampling, such that the larger the squared distance, the more likely a point gets chosen as the centroid.
5. Repeat *steps 3* and *4* until k centroids are chosen.
6. Proceed with the classic k-means algorithm.

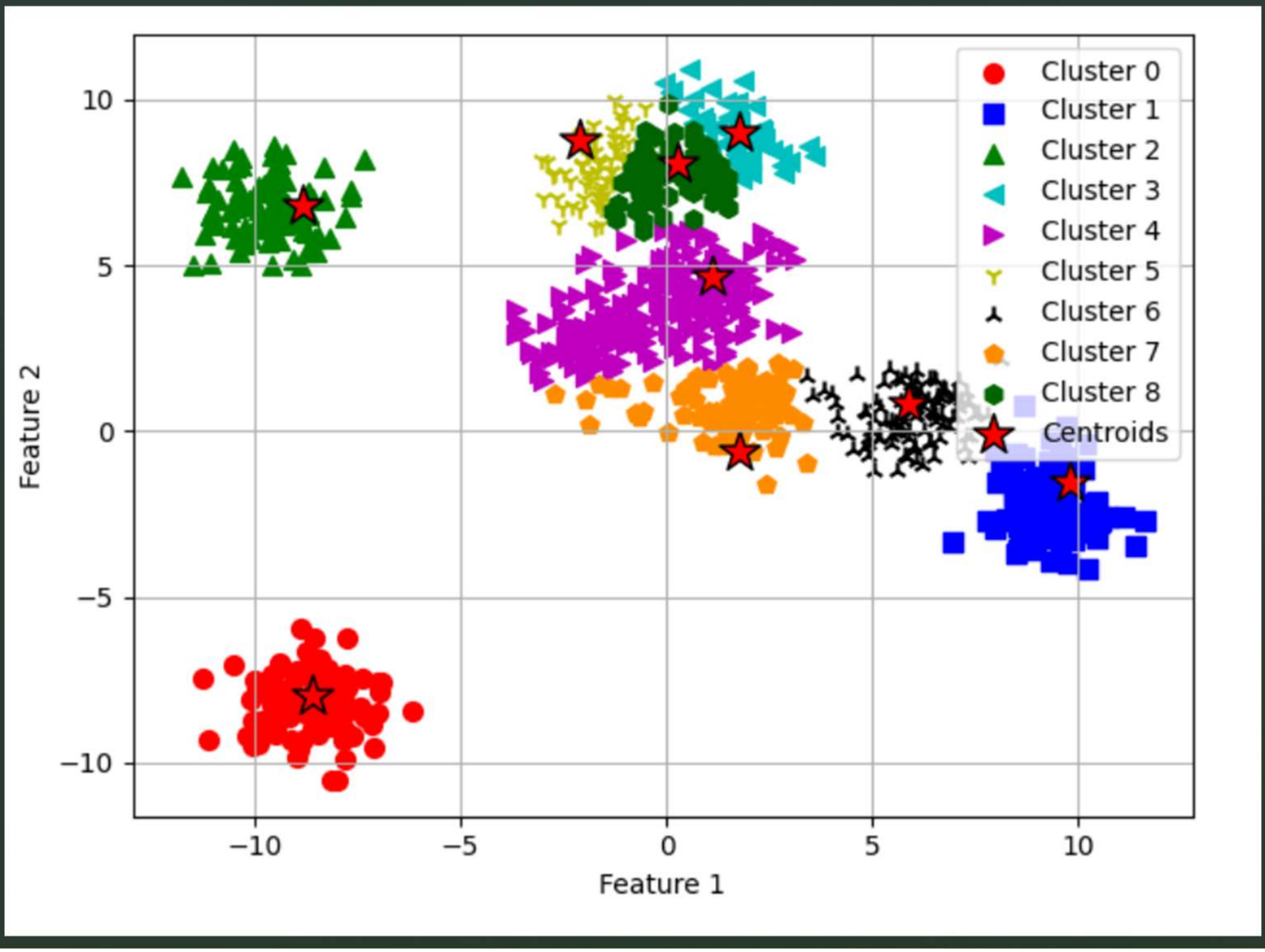
(from page 311 of [1])

Example : (k=9)



Cluster 0
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Cluster 6
Cluster 7
Cluster 8
Centroids
10
5
0
−5
−10
−10
−5
0
5
10
Feature 2
Feature 1

Example : (k=9)



Cluster 0
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Cluster 6
Cluster 7
Cluster 8
Centroids
Feature 1
Feature 2

P2 of homework 5 asks you to implement your own k-means++ algorithm

Hint : you may ask AI to
"implement k means++ algorithm class from scratch python"
and modify the code a little bit to suit the homework

Selected videos

CS229-2022 : Lecture 12 : K-means, GMM and EM
https://youtu.be/LMZZPneTcP4?si=yyNtXnx5Iwpj-urN

MIT Machine learning 6.036 : Lecture 13 : Clustering
https://youtu.be/BaZWcSq3IuI?si=jbusFioOt6eU2JBa

University of Tübingen Machine Learning I : 09 - Clustering and expectation-maximization
https://youtu.be/mU3GZaOoVDA?si=3j_tW6OXR6mFg3Me

references

- S. Raschka, Y. Liu and V. Mirjalili. Machine Learning with PyTorch and Scikit-Learn. Packt Publishing. 2022.
- A. Ng. CS229 Lecture Notes. Stanford University. 2023.
- D. Kobak. Machine Learning I. University of Tübingen. 2021.